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3.2.5. Numpy: Copying and Viewing Arrays

04:17

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the `original_array` and assigning it to `view_array`.
- Creating a copy of the `original_array` and assigning it to `copy_array`.

After completing these steps, observe how modifying the view affects the `original_array`, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:

Original array after modifying view: **<original_array >**

View array: **<view_array >**

- After modifying the copy:

Original array after modifying copy: **<original_array >**

Copy array: **<copy_array >**

```
import numpy as np
```

```
inputlist = list(map(int,input().split(" ")))
```

```
# Original array
```

```
original_array = np.array(inputlist)
```

```
# Create a view
```

```
view_array = original_array.view()
```

```
# Create a copy
```

```
copy_array = original_array.copy()
```

```
# Modify the view
```

```
view_array[0] = 99
```

```
print("Original array after modifying view:", original_array)
```

```
print("View array:", view_array)
```

```
# Modify the copy
```

```
copy_array[1] = 88
```

```
print("Original array after modifying copy:", original_array)
```

```
print("Copy array:", copy_array)
```

Average time
0.015 s
15.50 ms

Maximum time
0.021 s
21.00 ms

✓ 2 out of 2 shown test case(s) passed

✓ 2 out of 2 hidden test case(s) passed

✓ Test case 1 21 ms Debug

Expected output

Actual output

10 20 30 40 50 60 70 80	10 20 30 40 50 60 70 80
Original array after modifying view: [99 20 30 40 50 60 70 80]	Original array after modifying view: [99 20 30 40 50 60 70 80]
View array: [99 20 30 40 50 60 70 80]	View array: [99 20 30 40 50 60 70 80]
Original array after modifying copy: [99 20 30 40 50 60 70 80]	Original array after modifying copy: [99 20 30 40 50 60 70 80]
Copy array: [10 88 30 40 50 60 70 80]	Copy array: [10 88 30 40 50 60 70 80]

✓ Test case 2 21 ms